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GitHub Link	https://github.com/akshaya24032007/MLA0405-DeepLearning/tree/main/Assessment/CO3

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

**COURSE NAME: DEEP LEARNING
SLOT :D**

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

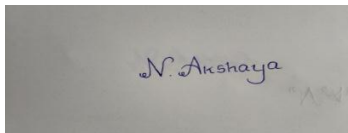
Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

I N.AKSHAYA(192425129) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student:N.AKSHAYA

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Answer:

Parameter	CNN	Fully Connected Neural Network
Architecture	Convolution, pooling and FC layers	Mainly fully connected layers
Feature extraction	Automatically extracts spatial features	Does not preserve spatial structure well
Parameters	Fewer due to parameter sharing	Very large for images
Computation	More efficient for images	Computationally expensive
Image handling	Preserves local spatial relationships	Image is usually flattened

Analysis: CNN filters detect local patterns such as edges, textures and shapes. Pooling reduces spatial dimensions and computational cost. FC layers perform final classification.

Conclusion: CNN is more suitable for X-ray classification because it preserves spatial information, requires fewer parameters and learns image features automatically.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Answer:

Layer	Main Function	Feature/Information
Convolution	Extracts features using filters	Edges, textures, shapes
Pooling	Reduces feature-map size	Retains important information
Activation	Adds non-linearity	Helps learn complex patterns
Fully Connected	Performs classification	Produces final class

Filters:

- **Early layers:** Learn simple features such as edges and corners.
- **Deeper layers:** Combine simple features to learn complex shapes and defect patterns.

Critical Evaluation: Convolution provides detailed feature extraction, while pooling reduces computation. Activation improves learning, and FC layers classify the extracted features.

Conclusion: The layers work together to identify industrial defects accurately and efficiently.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Answer:

Parameter	CNN Parameter Sharing	FC Independent Weights
Weights	Same filter reused across image	Separate weights
Parameters	Much fewer	Very high
Memory	Low	High
Computation	More efficient	More expensive
Feature detection	Detects features at different locations	Location-dependent
Translation handling	Better	Poorer

Analysis: A CNN uses the same convolutional filter at multiple image locations. Therefore, the network can detect the same feature wherever it appears.

Conclusion: Parameter sharing reduces memory and computational requirements while improving learning efficiency. It is especially important for thousands of high-resolution surveillance images.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Answer:

Parameter	AlexNet	ResNet
Depth	Relatively shallow	Much deeper
Organization	Conventional CNN	Uses residual/skip connections
Filters	Convolutional filters	Convolutional filters with residual blocks
Parameters	High	Depends on version

Parameter	AlexNet	ResNet
Feature extraction	Good	More powerful for complex features
Training	Easier for smaller depth	Skip connections ease deep training
Vanishing gradient	More limitation in deep extensions	Reduced using residual connections
Performance	Strong historical breakthrough	Generally better for complex recognition

Critical Evaluation: AlexNet was important in demonstrating the effectiveness of deep CNNs, but very deep networks can suffer from training difficulties. ResNet addresses this using **skip connections**, allowing information and gradients to flow through deeper networks.

Recommendation: ResNet is more suitable for complex image recognition because it supports deeper architectures and stronger hierarchical feature learning.

Insight: The major advantage of ResNet is not simply having more layers, but making very deep networks easier to train.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization** and a **CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Answer:

Parameter	Non-Regularized CNN	Regularized CNN
Overfitting	High	Reduced
Generalization	Poor	Better
Training accuracy	May be very high	Usually more balanced
Stability	Can be unstable	More stable
Unseen images	Poor performance	Better performance

Regularization Techniques

- **Dropout:** Randomly disables neurons during training, reducing dependence on particular neurons.
- **Data Augmentation:** Creates variations of training images such as rotations or flips, increasing data diversity.
- **Batch Normalization:** Normalizes intermediate activations and can improve training stability.

Critical Evaluation: Since the model has high training accuracy but poor unseen-image performance, **overfitting** is the main problem.

Recommendation: Use **data augmentation + dropout + batch normalization**. Data augmentation improves diversity, dropout controls overfitting, and batch normalization improves training stability.

Conclusion: A regularized CNN provides better generalization and is more reliable for real-world plant disease classification.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand